

JOB 2 : Assembling and Disassembling of Laptop



OBJECTIVE : Disassembly, Component Identification, and Reassembly of a Laptop Computer

a) Disassemble a Laptop

Step 1. Power off the laptop and disconnect all cables and peripherals.

Step 2. Remove the battery if it is accessible and easily removable.

Step 3. Locate and remove all screws from the bottom of the laptop using the appropriate screwdriver.

Step 4. Use a Uspudger to carefully pry open the laptop casing, starting from the edges and working your way around.

Step 5. Once the casing is open, carefully disconnect any cables or ribbons connecting the keyboard, trackpad, and other components to the motherboard.

Step 6. Remove the keyboard and trackpad by unscrewing them from the casing and gently lifting them out.

Step 7. Unscrew and remove the motherboard, taking note of any remaining connections or screws holding it in place.

Step 8. If necessary, remove and replace any components such as the hard drive, RAM, or CPU.

Step 9. Reassemble the laptop by following these steps in reverse, making sure to reconnect all cables and screws securely.

Step 10. Power on the laptop to ensure everything is functioning properly.

b) Identification of the components individually :

- 1.CPU- (Brain of Computer): AMD A6 5300M
- 2.RAM : ADATA 8 GB DDR3L RAM
- 3.Cooling Fan
- 4.Motherboard-Connects and Transmits Signals from Peripherals
- 5.Hard Disk drive & Solid State drive : 256 GB
- 6.Battery
- 7.TouchPad
- 8.Speakers
- 9.KeyBoard



c) Description of each identified component along with the image of the component.

CPU

CPU is the principal part of any digital computer system, generally composed of the main memory , control unit, and arithmetic-logic unit. It constitutes the physical heart of the entire computer system; to it is linked various peripheral equipment, including input/output devices and auxiliary storage units. In modern computers, the CPU is contained on an integrated circuit chip called a microprocessor .

The control unit of the central processing unit regulates and integrates the operations of the computer. It selects and retrieves instructions from the main memory in proper sequence and interprets them so as to activate the other functional elements of the



system at the appropriate moment to perform their respective operations. All input data are transferred via the main memory to the arithmetic-logic unit for processing, which involves the four basic arithmetic functions (i.e., addition, subtraction, multiplication, and division) and certain logic operations such as the comparing of data and the selection of the desired problem-solving procedure or a viable alternative based on predetermined decision criteria .

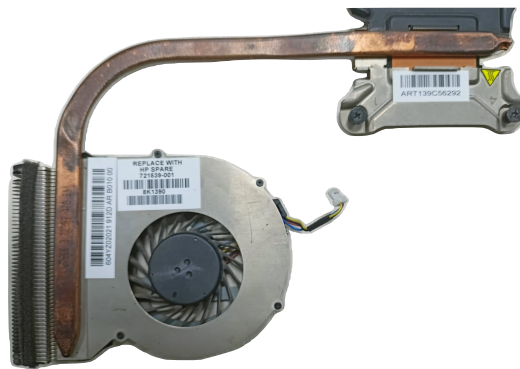
RAM

RAM is the Computer's main memory in which specific contents can be accessed (read or written) directly by the central processing unit in a very short time regardless of the sequence (and hence location) in which they were recorded. Two types of memory are possible with random-access circuits: static RAM (SRAM) and dynamic RAM (DRAM). A single memory chip is made up of several million memory cells. In a SRAM chip each memory cell stores a binary digit (1 or 0) for as long as power is supplied. In a DRAM chip the charge on individual memory cells must be refreshed periodically in order to retain

data . Because it has fewer components, DRAM requires less chip area than SRAM. Hence, a DRAM chip can hold more memory, though its access time is slower.

Cooling Fan

A CPU fan, or cooling fan, is an active cooling device that blows air over the heat sink to accelerate the heat dissipation process. It is typically mounted on top of or alongside the heat sink. The fan helps to maintain a lower temperature around the CPU by continuously moving air, preventing the CPU from overheating. The speed of the fan can be controlled dynamically based on the CPU temperature to optimise cooling efficiency.



Motherboard

A laptop motherboard is the primary printed circuit board (PCB) inside the laptop that contains the components that make the computer work. The motherboard, the central hub of a laptop, is a complex and integral component, hosting a variety of elements that drive the device's functionality.



Components of a Motherboard

Mouse and Keyboard Connectors: These connectors are like plugs to which we attach our mouse and keyboard. So they can effectively communicate with the computer system.

Universal Serial Bus (USB) USB ports connect a wide range of peripheral devices directly to the motherboard, including external hard drives and printers.

RAM (Random Access Memory): RAM provides temporary storage for data like the computer's short-term memory. It allows the CPU to access data quickly, which helps improve overall system performance.

CPU (Central Processing Unit) :The CPU is like the computer's brain. It performs calculations and is responsible for executing instructions.

Basic Input/Output System (BIOS) :The BIOS is firmware that initialises hardware components during startup. It also provides low-level control of certain system functions.

Adapter Cards and Expansion Slots :These slots allow adding components to the motherboard, like better graphics cards, sound cards, or network cards, to increase the computer's capabilities.

Chipsets :Chipsets manage data flow between the RAM, CPU, and other peripheral devices. They help different parts to communicate with each other and work together.

Storage Devices:Storage devices are like hard and solid-state drives connected to the motherboard. They store your operating system, data, and programs.

CMOS Battery: The CMOS battery is like a small battery that keeps some necessary settings safe, even when the computer is off.

Power Connector This connector supplies power from the computer's power supply unit (PSU) to the motherboard. It enables the motherboard and its components to operate.

Hard Disk Drive & Solid State Drive

An HDD uses magnetism, which allows you to store data on a rotating platter. It has a read/write head that floats above the spinning platter for Reading and Writing of the data. The faster the platter spins, the quicker an HDD can perform. HDD also consists of an I/O controller and firmware, which tells the hardware what to do and communicates with the remaining system. The full form of HDD is Hard Disk Drive

Solid State Drive (SSD) is a non-volatile storage device that stores and retrieves data constantly on solid-state flash memory. However, this data is stored on interconnected flash memory chips instead of platters, which makes them faster than HDDs. It provides better performance compared to HDD. In terms of speed, SSDs are much faster than HDDs because SSDs do not have electrical circuitry and moving parts as HDD. A typical 7200 RPM HDD can deliver read/write speed of 80 to 160 MB/s while a typical SSD can deliver read/write speed of 200 to 550 MB/s which makes it an ideal choice for better performance.

HDD

Storage: An HDD is like a digital storage room for your computer. It can store a lot of data, including your operating system, applications, documents, pictures, and videos.

Mechanical Parts: Unlike newer technologies such as Solid State Drives (SSDs), HDDs have moving parts. Inside the hard drive, there are spinning disks (platters) and a read/write head. The data is stored on these spinning disks, and the read/write head moves back and forth to access or modify the data.

Reading and Writing: When you open a file or run a program, the read/write head finds and reads the necessary data from the spinning disks. When you save a file, it writes the data back to the disks.

Capacity: HDDs come in various sizes, commonly ranging from a few hundred gigabytes (GB) to several terabytes (TB) of storage space. The larger the capacity, the more data it can store. **Speed:** In



general, HDDs are slower than newer SSDs. The speed is affected by the time it takes for the spinning disks to rotate and the read/write head to move. However, they are still widely used for storing large amounts of data at a more affordable cost compared to SSDs.

SSD

Speed: SSDs are much faster than HDDs. They can access and retrieve data quickly, which results in

faster boot times, faster application loading, and overall improved system responsiveness.

Durability: Since SSDs have no moving parts (unlike the spinning disks in HDDs), they are more durable and less prone to mechanical failure. This makes them a more reliable choice for storing important data. **Size and Weight:** SSDs are generally smaller and lighter than HDDs, making them ideal for laptops and

other portable devices where space and weight are crucial factors.

Energy Efficiency: SSDs consume less power than HDDs, contributing to longer battery life in laptops and lower power bills for desktop computers.

Battery

A laptop battery is a rechargeable power source that provides the necessary energy for a laptop to operate without being connected to a direct power source.

It is used to provide power backup in order to operate a laptop when AC mains power is not available. The common laptop battery voltages are 10.8V, 11.1V or 14.8V.

The common laptop battery types include lithium ion (i.e., Li-ion), nickel cadmium (i.e., NiCad) and nickel metal hydride (i.e., NiMH). Out of these Li-ion is being most commonly found in latest laptops. They are made using various combinations of cells viz. 3 cells or 4 cells etc.

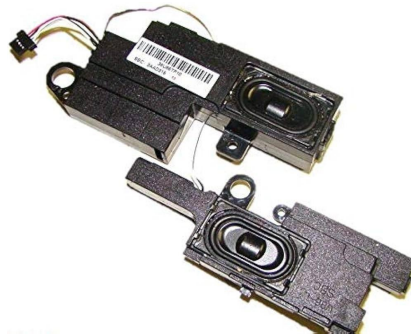
TouchPad

The touchpad or trackpad uses a tactile sensor on a specialised surface. The touchpad is a pointing device which translates motion and position of fingers on the surface to relative cursor position on the screen of the operating system installed in the laptop. It is the most common feature of all the laptops and used as a substitute for a mouse where desk space is limited.



Speaker

Speakers are used to produce sound. Like external speakers, laptop speakers also allow users to control volume up or down as per need. It is available in power ratings of 2.6 watt or 5 watts.



KeyBoard

It is most common input device used to provide commands or data to the laptop or computer. There are different types of keyboards based on keys used and their layout arrangements. The common types are QWERTY, QWERTZ, AZERTY and HCESAR. Various keys found on keyboards include standard typewriter keys, function keys (F1, F2, F3, ..., F12) special purpose keys (Ctrl, Del, Ins, Alt) , cursor movement keys, numeric keys.

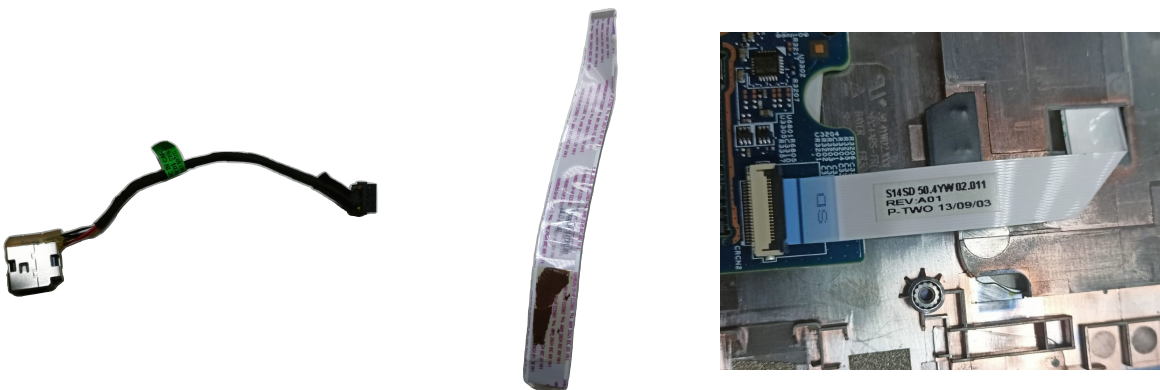
Connectors:



Internal Connectors are located inside a computer case. PC internal cables make the core of a computer. There are basically two main types of connectors found in the computer internally: socket connectors and power connectors.

Power connectors are used to supply and distribute power to internal devices inside the computer. They normally use the friction of the connector bodies to stay in place.

Socket connectors are designed for use with flat ribbon cable, and are generally used to transfer data among devices.



d) Assemble the Laptop back :

Step 1. Begin reassembly by connecting components to the motherboard.

Step 2. Secure the cooling fan, wireless card, and other components.

Step 3. If separated, carefully reattach the display assembly to the laptop's base.

Step 4 . Place the bottom cover back on and secure it with screws.

Step 5. Place RAM modules back into their slots and secure the hard drive.

Step 6. If applicable, reconnect the laptop battery.

Step 7. Power on the laptop and ensure that it boots up successfully.

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JOB-3

AIM :

Preparation of a network cable, commonly known as an Ethernet cable, for creating reliable connections in networking setups.

MATERIAL REQUIRED:

•Ethernet cable (CAT5e, CAT6, or CAT6a)

- An Ethernet cable is a type of network cable commonly used to connect devices in a local area network (LAN) for the purpose of transmitting data.
- It is the most common type of wired network connection and is widely used in homes, offices, and data centers for networking various devices such as computers, printers, routers, switches, and more.



- **RJ45 connectors (compatible with the cable type)**
 - RJ45 stands for Registered Jack 45 and is the most commonly used connector in wired networks. The jacks are mainly used to connect to the Local Area Network (LAN). It was earlier devised for telephones but is now widely used in Ethernet Networking. The 45 in RJ45 basically stands for the listing number.



- **Crimping tool**
 - A crimping tool is the tool used to deform the material and create the connection. Crimping is commonly used in electrical work, to attach wires together or wire to other connectors.



- **Cable cutter/stripper**
- **Cable tester (optional but recommended)**
- **Marker pen (for labeling, optional)**



STEPS TO MAKE NETWORK CABLE:

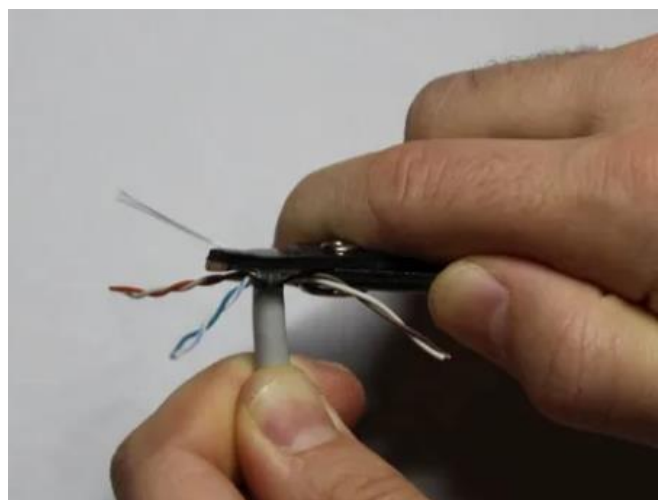
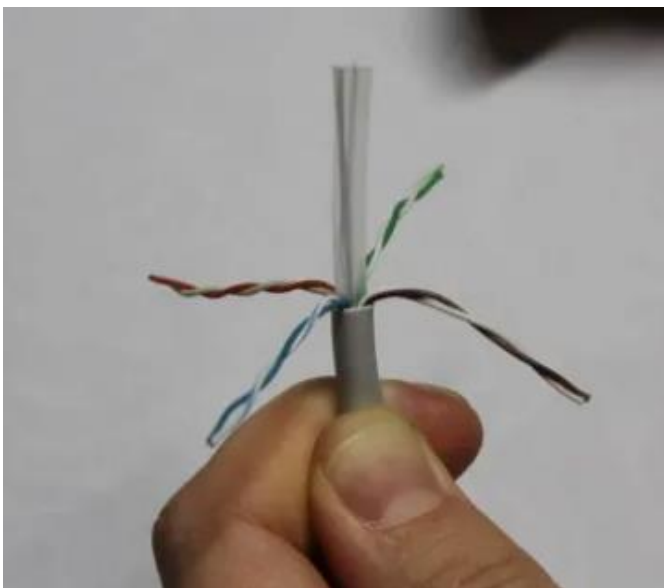
1. Strip the Cable Ends:

We make a cut from about 2,5 cm from both ends of the cable and peel off the PCV cable sheath that is located here.



2. Seperate the Pairs and Remove the Protector in the Middle:

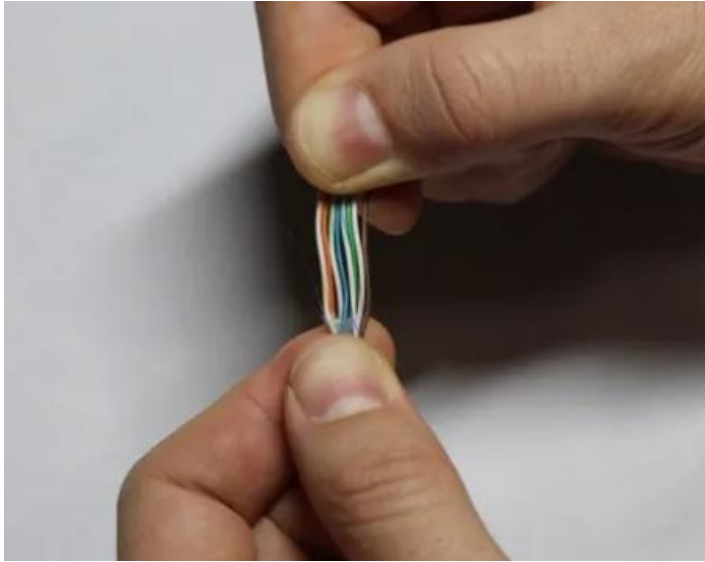
If you are using a Cat6 cable, we have a plastic cable organizer in the middle of the cables and we cut the nylon cable from the bottom. Cat5E cable does not have such a regulator.



3. Order the Pairs

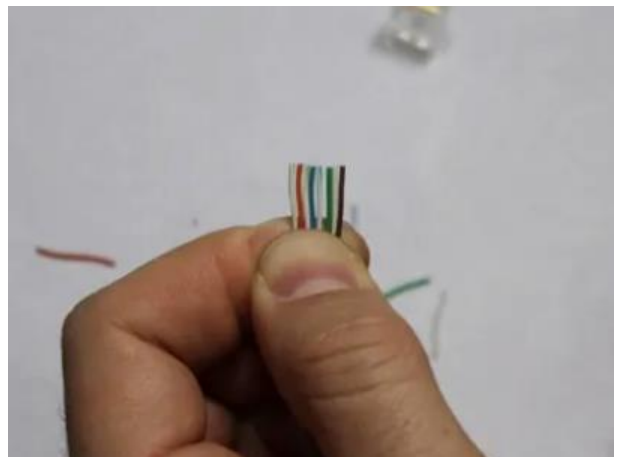
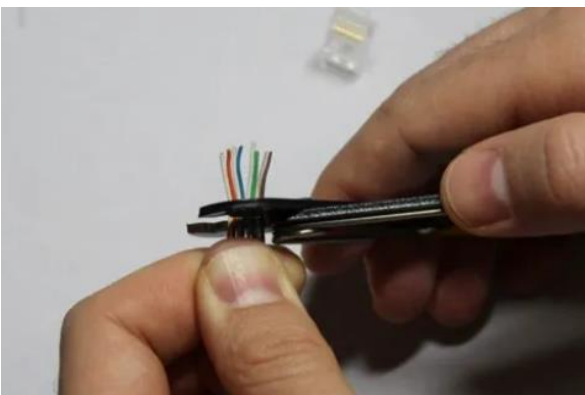
Separate pairs from each other and order them according to color code.

1. Orange-White
2. Orange
3. Green-White
4. Blue
5. Blue-White
6. Green
7. Brown-White
8. Brown



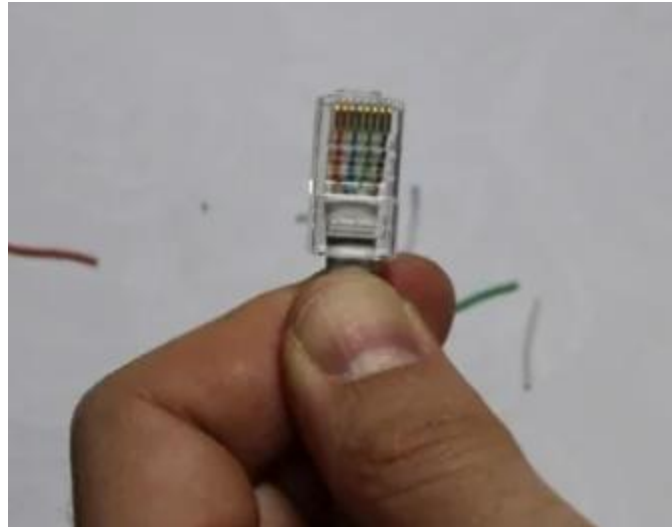
4. Shorten the Pairs:

While keeping the cables together in the proper sequence, we cut the cable lengths so that they are about 1.2 cm long. To determine the length as an estimate, we can make a measurement with connector and cut it accordingly.



5. Insert Cables to the Connector:

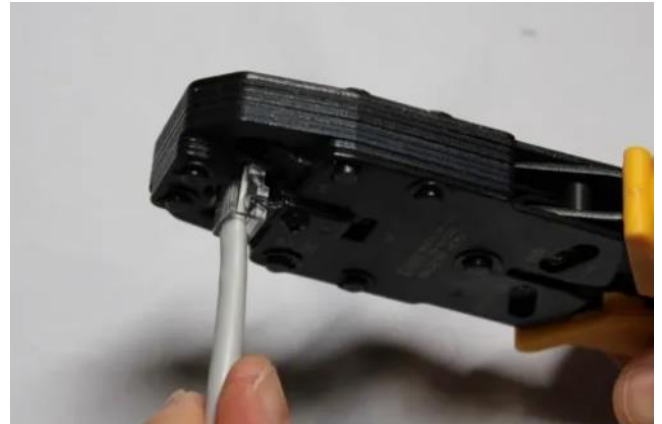
We steer the cables properly while you hold them, in while you need to slowly insert them into the RJ45 connector in the proper sequence. Please be careful, don't lose the cable order when inserting cables. After insertion you have to ensure that cables are inserted with correct order.



6. Insert to the Crimping tool and Press:

We hold the RJ45 connector and cable carefully and place it in the RJ45 connector tightening area of the crimping tool.

We tighten the crimping tool to ensure that the pins on the connector are jammed and the cable strand is bitten by them.



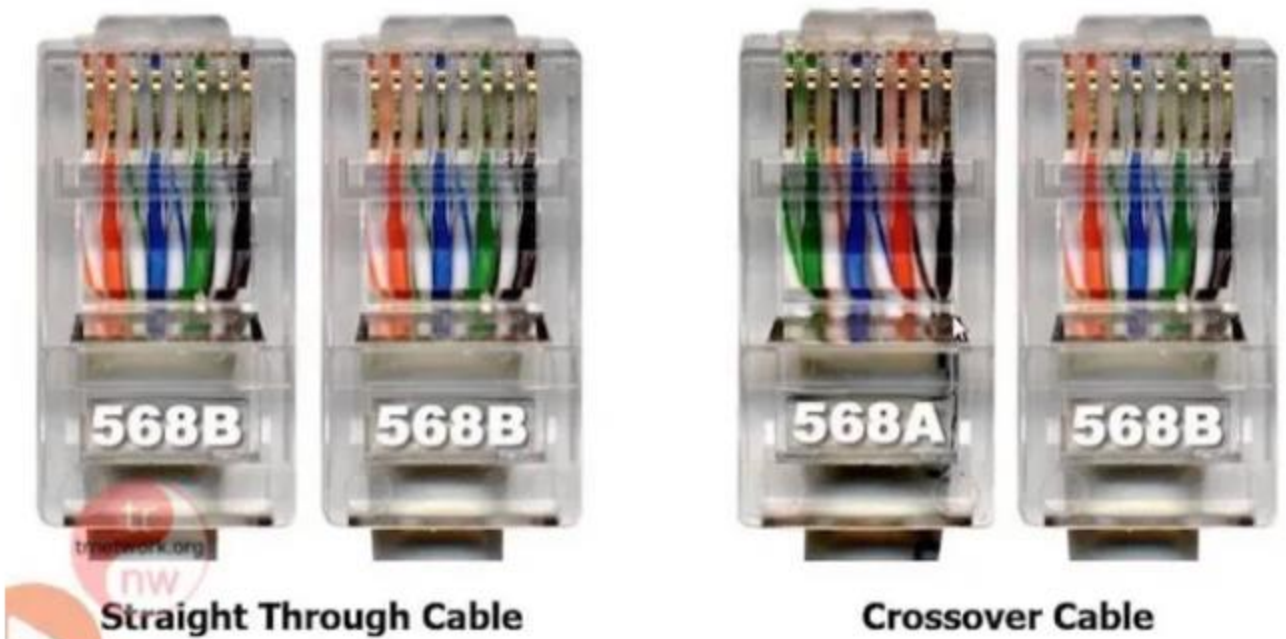
7. Check:

We visually check that the pins and the connector is fully seated.



8. Do the Same for The Other End:

We repeat the same process for the other end. If cross cable is to be made, we must remember that the other end must be screwed in accordance with T568A standard.



CHECK THE CONNECTION

You can check the connection of Ethernet between two laptops by following these steps:

Connect the Ethernet cable:

First, ensure that both laptops have Ethernet ports. Then, connect one end of the Ethernet cable to the Ethernet port of the first laptop and the other end to the Ethernet port of the second laptop.

Check network settings:

On both laptops, navigate to the network settings. This can usually be found in the Control Panel or Settings menu, depending on the operating system.

Ensure that Ethernet or LAN connection is enabled.

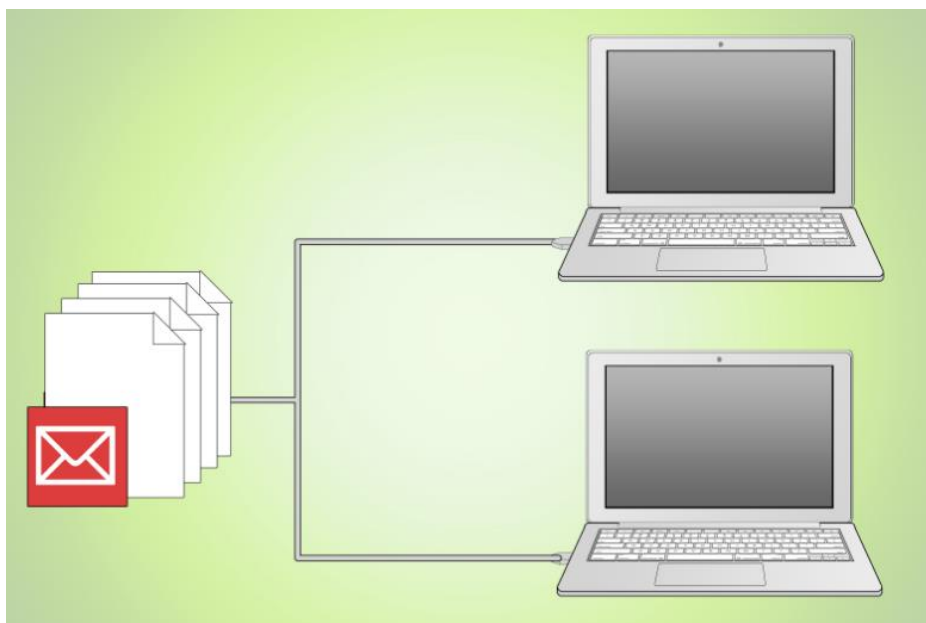
If both laptops are set to obtain IP addresses automatically (via DHCP), they should assign IP addresses to each other automatically once the connection is established.

Verify connection:

Open a command prompt or terminal on one of the laptops.

Use the ping command to check connectivity to the other laptop. For example, if the IP address of the second laptop is 192.168.1.2, you would type ping 192.168.1.2 and press Enter.

If the connection is successful, you should see responses from the second laptop indicating that the packets are being received.



CONCLUSION

This workshop provided valuable hands-on experience in the preparation of Ethernet cables, essential components in computer networking. Through meticulous attention to detail and adherence to industry standards, participants gained proficiency in the proper termination and alignment of cables, ensuring reliable connectivity. By mastering these skills, attendees are better equipped to deploy and troubleshoot Ethernet connections effectively, contributing to the seamless operation of modern networking infrastructures.

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Job-4: OS Installation

1.Bootable Pen drive with windows

1.Download windows tools through its official web site

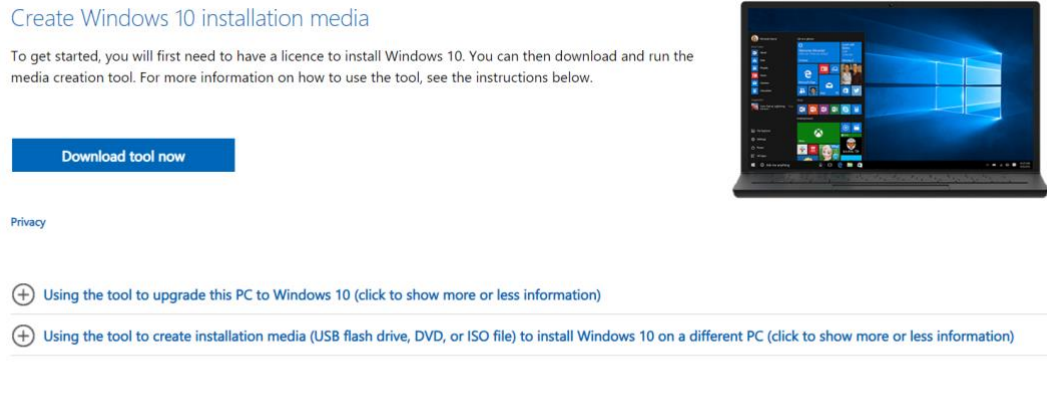


Fig.1

2. As the file is downloaded, a dialogue box opens giving us a choice

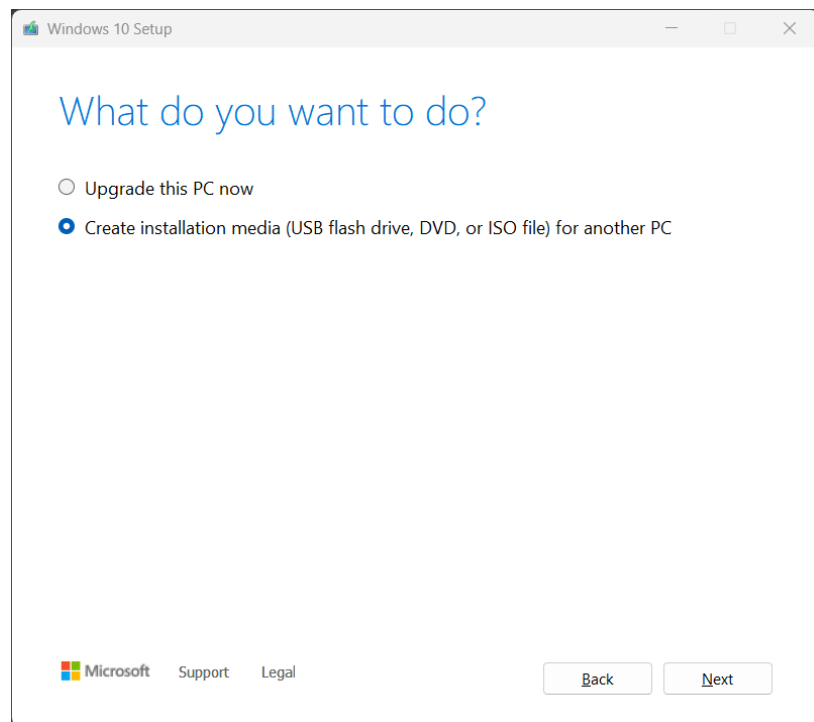


Fig.2

3. Choose which type of media you are willing to use, we will be saving it in a bootable pen-drive

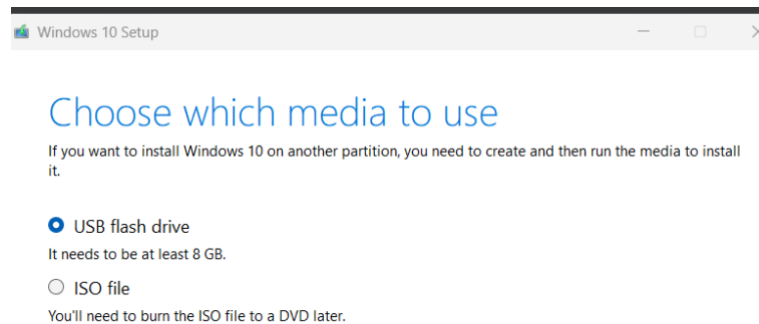


Fig.3

4. Our bootable pen drive is ready

Click on setup option.

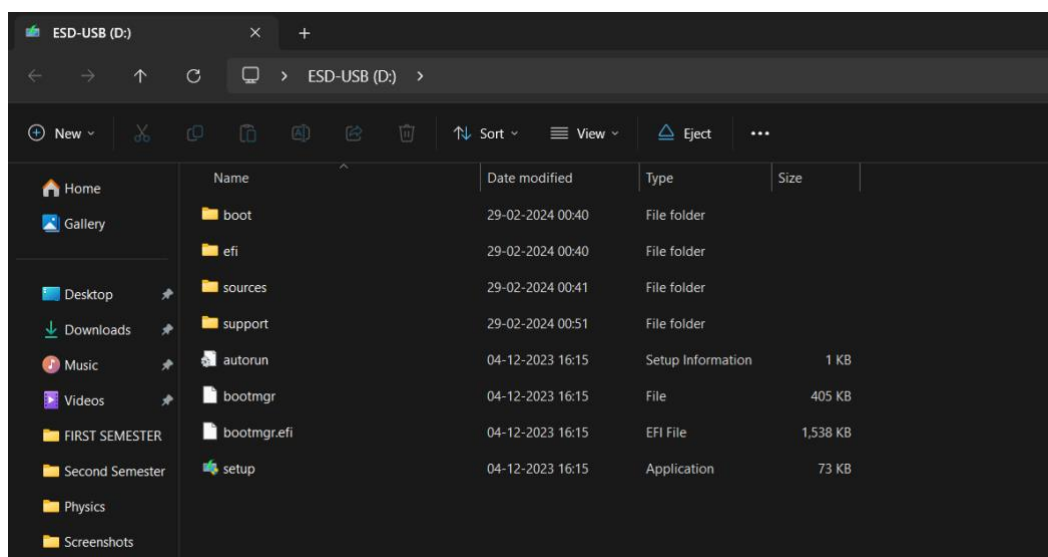


Fig.

5. Using the pen-drive to run windows on our computer

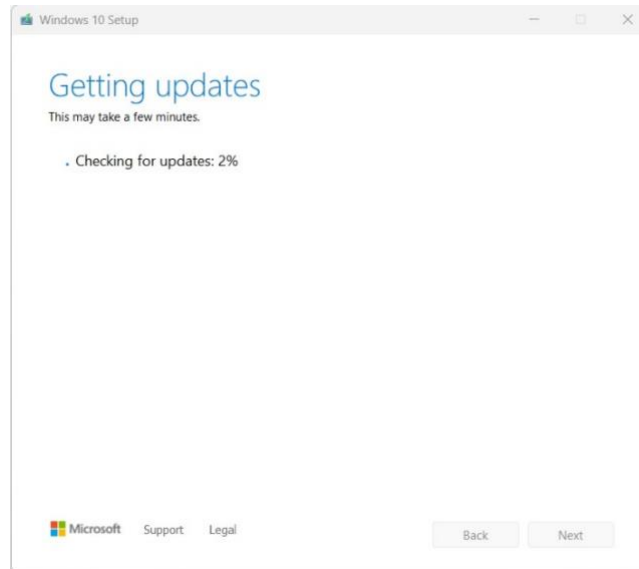


Fig.5

Set-up: Setting up our windows installation

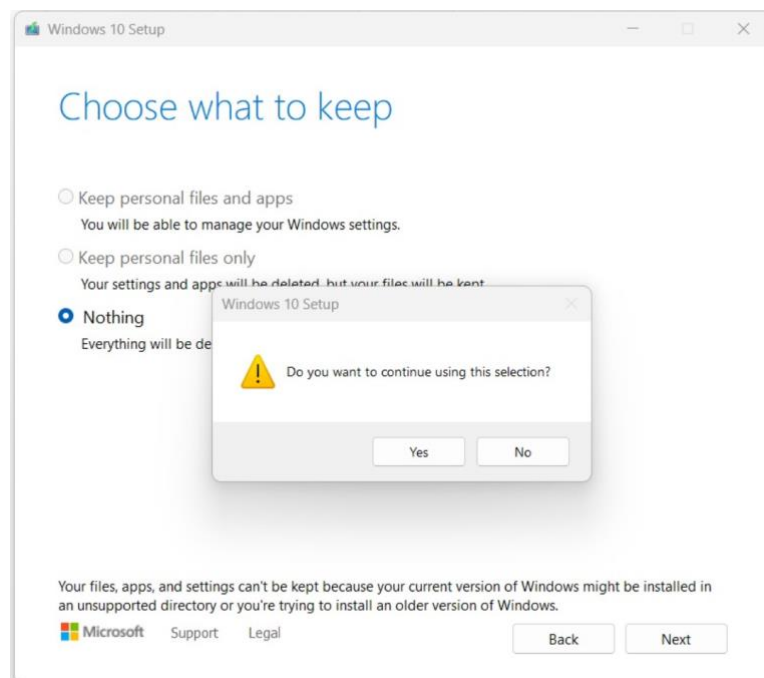


Fig.6

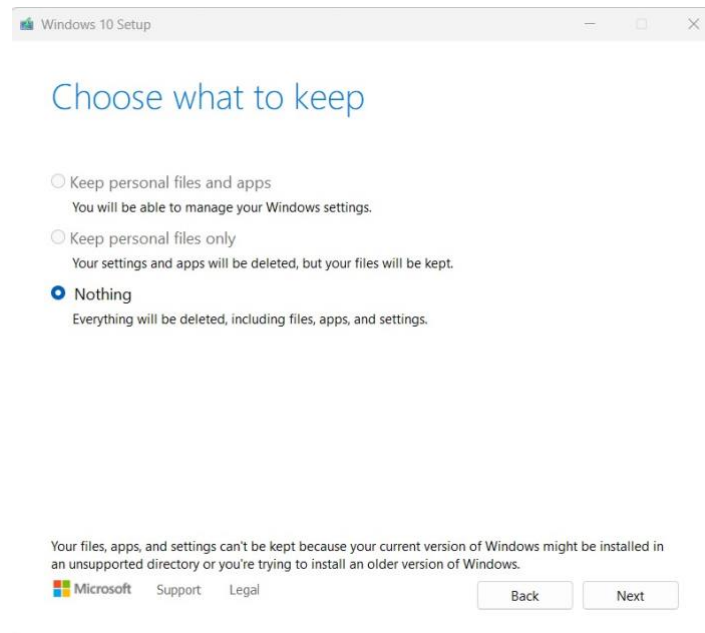


Fig.7

2.. Installing Ubuntu on your PC

Objective To install OS Software and get acquainted with their workings

Materials Required: Laptop , Internet Connection

Procedure

A. UBUNTU

Visit the Ubuntu Website

Go to the official Ubuntu website at

<https://ubuntu.com/download/desktop>.

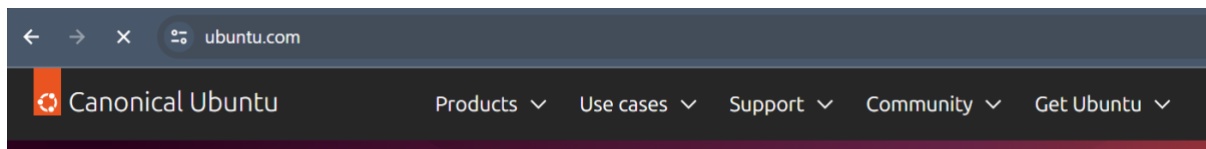


Fig.8

Choose Your Version

On the Ubuntu Desktop download page, you will see the latest version available. If

you want the LTS (Long Term Support) version, it's usually prominently displayed.

Click on the version you prefer. For example, "Download Ubuntu 20.04 LTS" or

whichever version is the latest at the time.

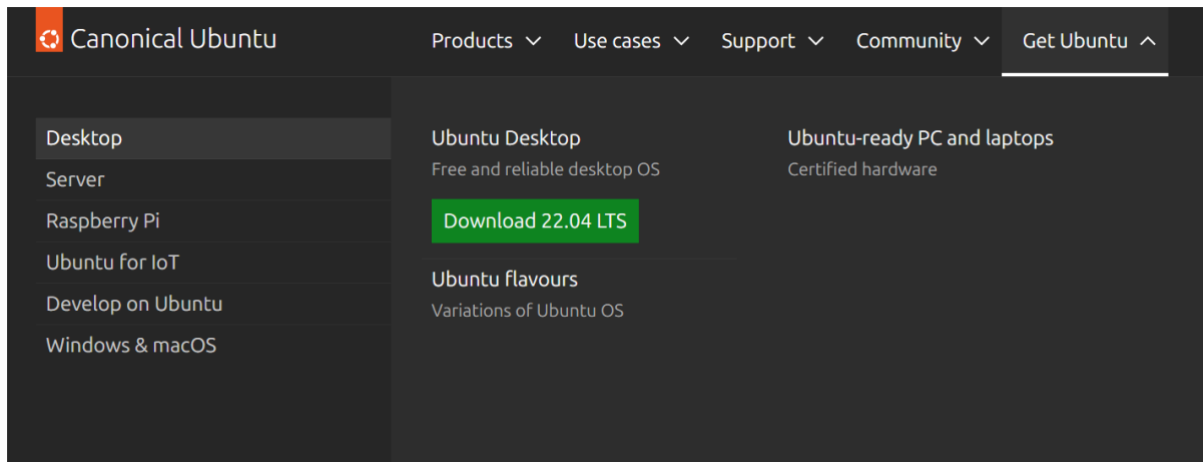


Fig.9

Select 32-bit or 64-bit

You'll then be asked to choose between 32-bit and 64-bit. Generally, 64-bit is

recommended if your system supports it, as it allows you to use more memory and

provides better performance. Select the version you want and click "Download".

Download the ISO File

After clicking "Download", you'll be redirected to a page where you can choose to

donate (optional) or simply download the ISO file. Click on the "Download" button to

start the download.

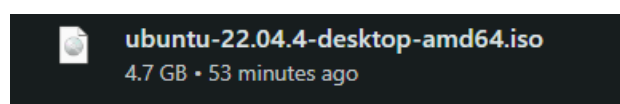


Fig.10

Save the ISO File

You will be prompted to save the ISO file. Choose a location on your computer where

you want to save the file and click "Save".

Verify the Download (Optional but Recommended)

It's good practice to verify the integrity of the downloaded file to ensure it hasn't been

corrupted during the download process.

B. VirtualBox

Visit the VirtualBox site

Go to the official VirtualBox site <https://www.virtualbox.org/>



Fig.10

Download the appropriate version

On the official VirtualBox website you will get options to download VirtualBox

according to your OS system.



Fig.11

Run the installer

Once the download is complete, run the installer (typically a file with a .exe extension). You might see a security warning; proceed by clicking "Yes" or "Run" to allow the installer to run.

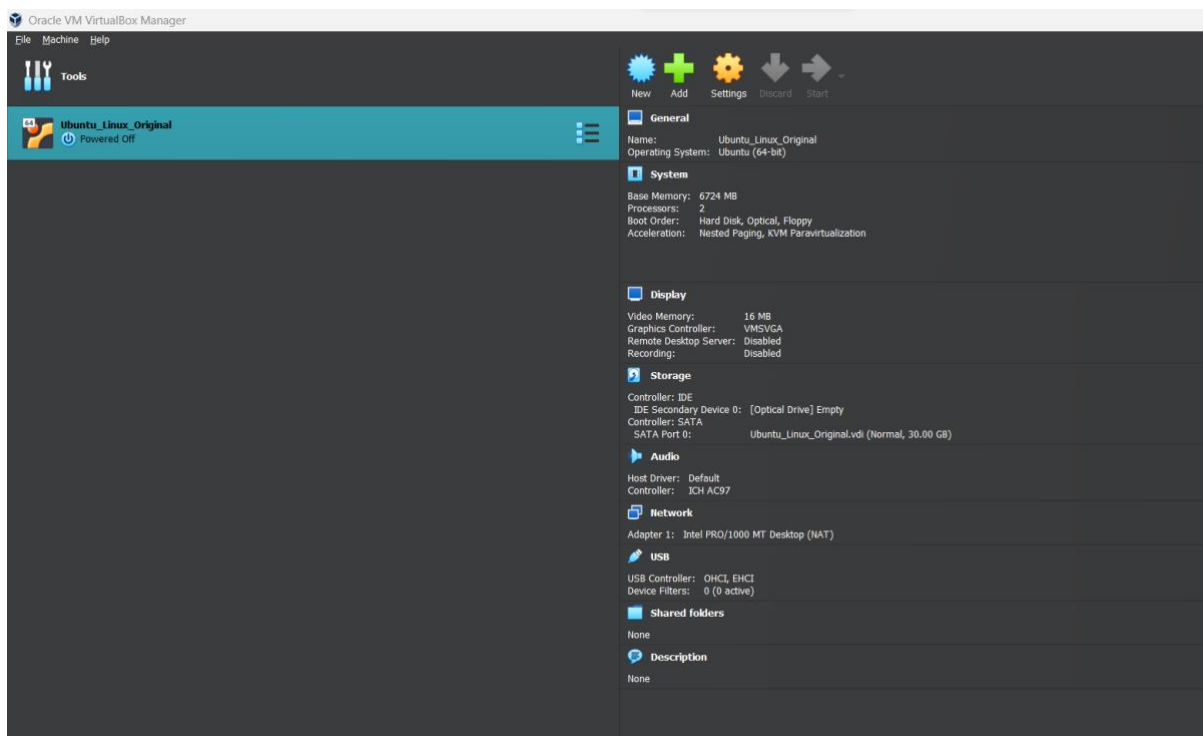


Fig.12

- Our Linux admin window opens and asks for the password

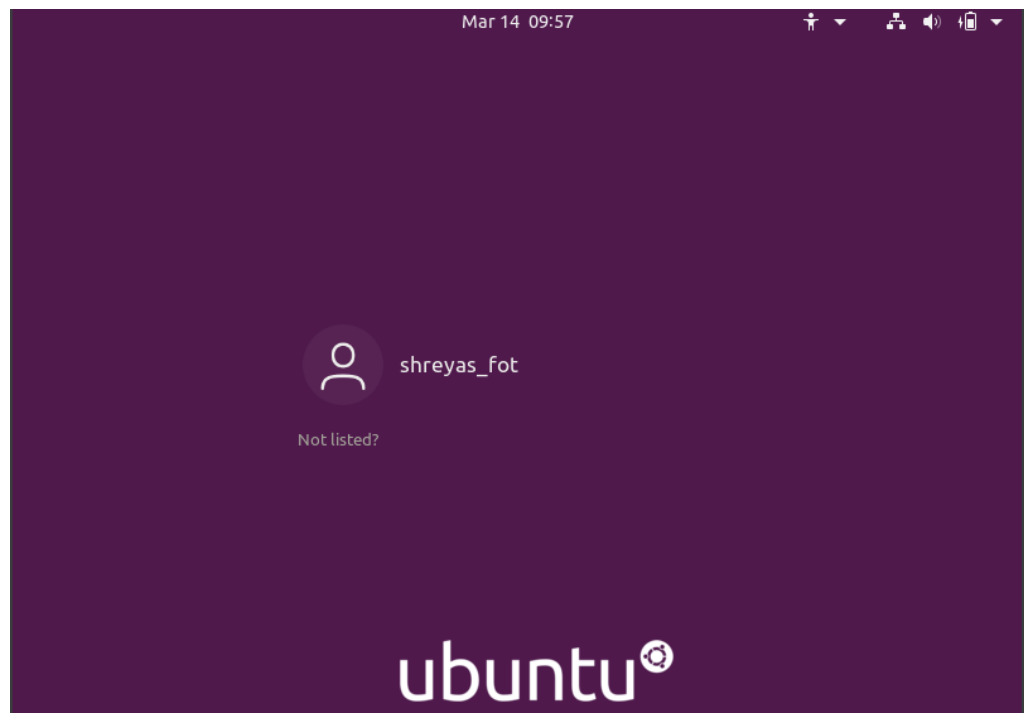


Fig.13

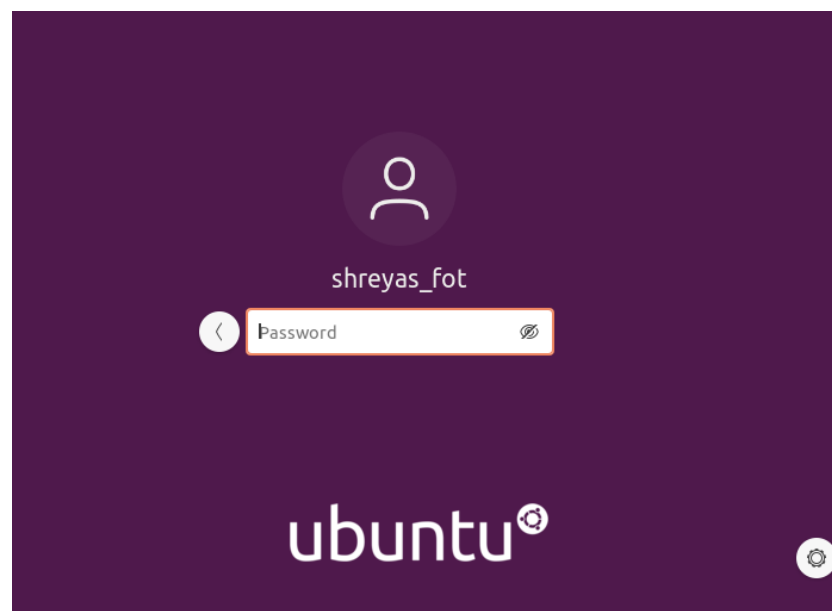


Fig.14

- After we enter the password our ubuntu gui desktop is shown to us

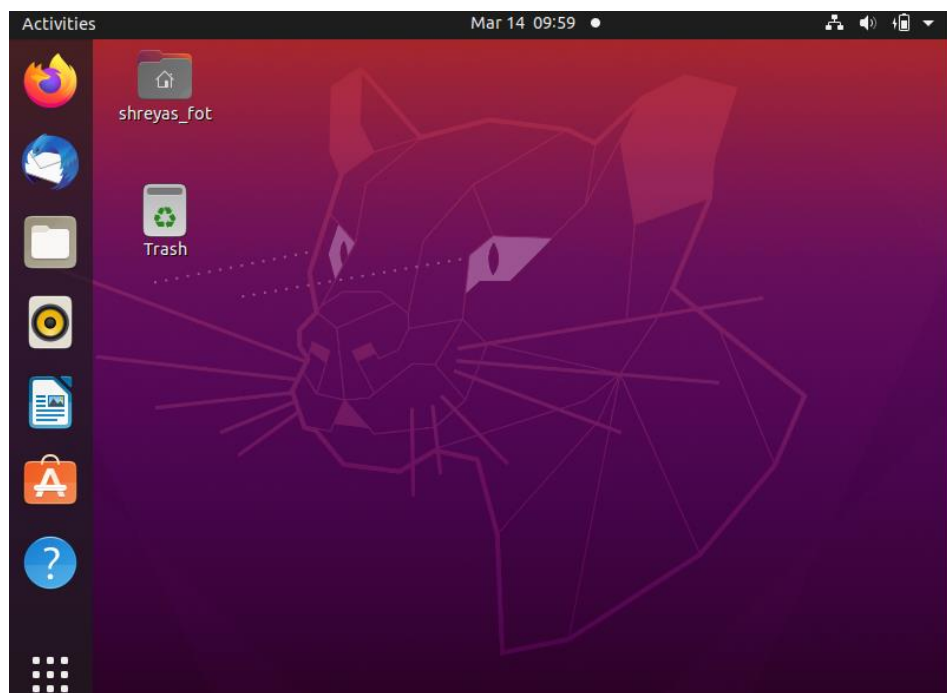


Fig.15

3. Running applications on Ubuntu

1) Downloading VLC in our ubuntu Linux through GUI interface

- First we click on app store in ubuntu
- Click on install button

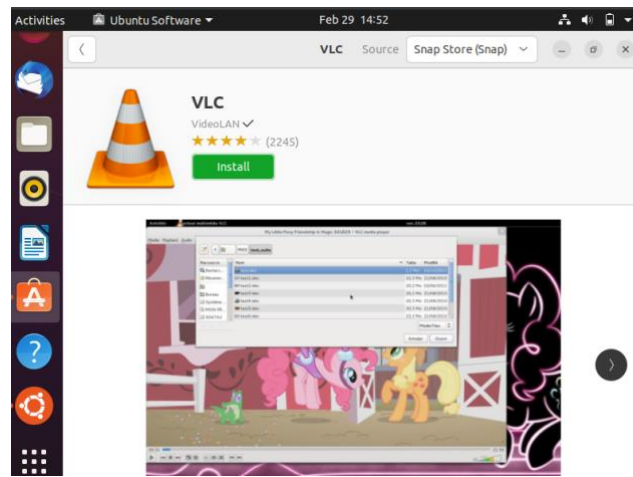


Fig.16

2. Vlc is running –

- Right click on VLC icon
- Click on open

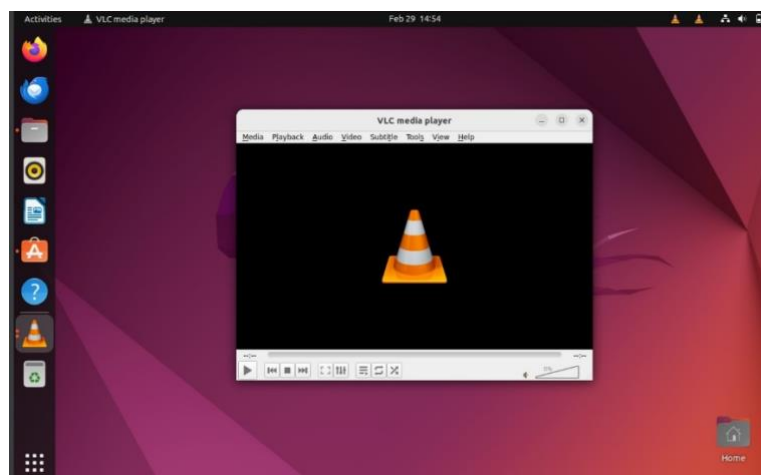


Fig.17

2) Downloading Chrome Browser

1.Open Linux Terminal and write the following command in the window

- `wget https://dl.google.com/linux/direct/google-chrome-stable_current_amd64.deb`

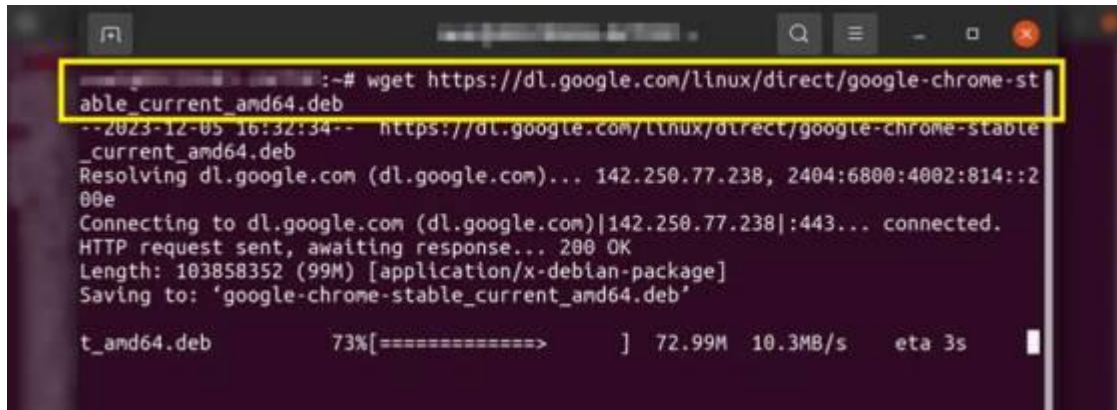


Fig.18

- `$sudo dpkg -i google-chrome-stable_current_amd64.deb`



Fig.19

- `sudo apt-get install -f`

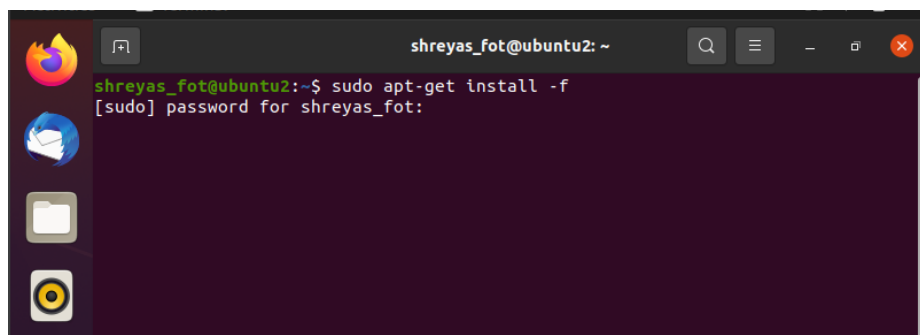


Fig.20

2.Type the password and chrome downloading process will begin

```
Connecting to dl.google.com (dl.google.com)|172.217.24.238|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 58545656 (56M) [application/x-debian-package]  
Saving to: 'google-chrome-stable_current_amd64.deb'  
  
google-chrome-stabl 100%[=====] 55.83M 611KB/s in 1m 46s  
  
2019-04-24 11:16:57 (540 KB/s) - 'google-chrome-stable_current_amd64.deb' saved  
[58545656/58545656]
```

Fig.21

3.Google chrome is installed

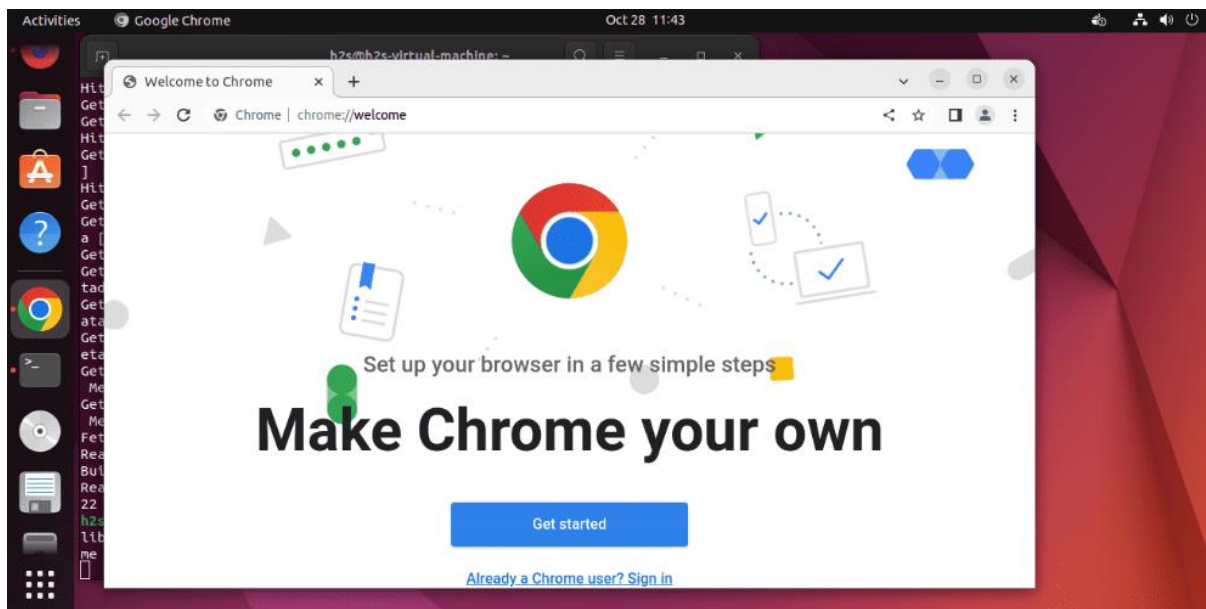
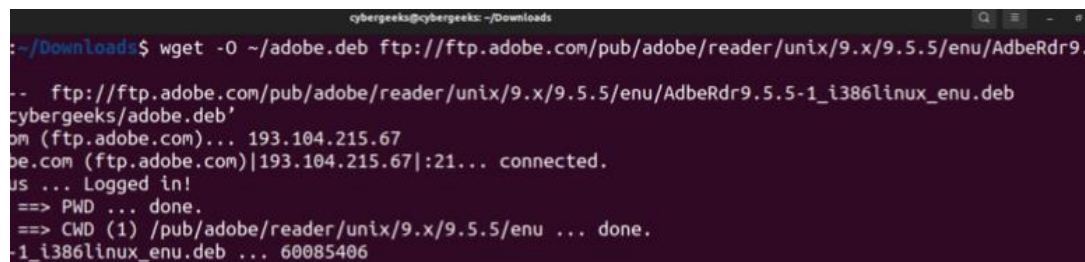


Fig.22

3) Downloading Adobe pdf Reader

1. Open Linux Terminal and write the following command in the window

- `wget -O ~/adobe.deb ftp://ftp.adobe.com/pub/adobe/reader/unix/9.x/9.5.5/enu/AdbeRdr9.5.5-`

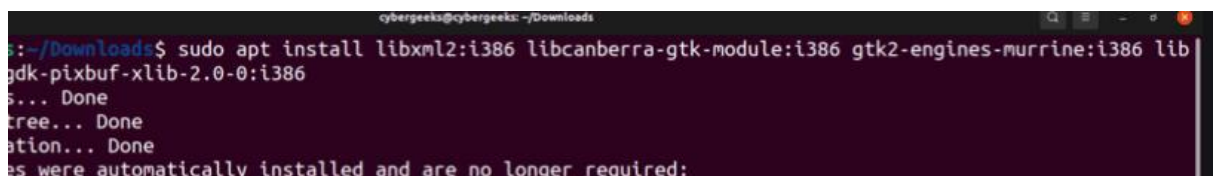


```
cybergeeks@cybergeeks: ~/Downloads
~/Downloads$ wget -O ~/adobe.deb ftp://ftp.adobe.com/pub/adobe/reader/unix/9.x/9.5.5/enu/AdbeRdr9.5.5-1_i386linux_enu.deb
-- ftp://ftp.adobe.com/pub/adobe/reader/unix/9.x/9.5.5/enu/AdbeRdr9.5.5-1_i386linux_enu.deb
cybergeeks/adobe.deb'
on (ftp.adobe.com)... 193.104.215.67
be.com (ftp.adobe.com)|193.104.215.67|:21... connected.
us ... Logged in!
==> PWD ... done.
==> CWD (1) /pub/adobe/reader/unix/9.x/9.5.5/enu ... done.
_1_i386linux_enu.deb ... 60085406
```

Fig.23

→ This installs adobe deb package on our linux virtual machine

- `sudo apt install libxml2:i386 libcansberra-gtk-module:i386 gtk2-engines-murrine:i386 libatk-adaptor:i386 libgdk-pixbuf-xlib-2.0-0:i386`

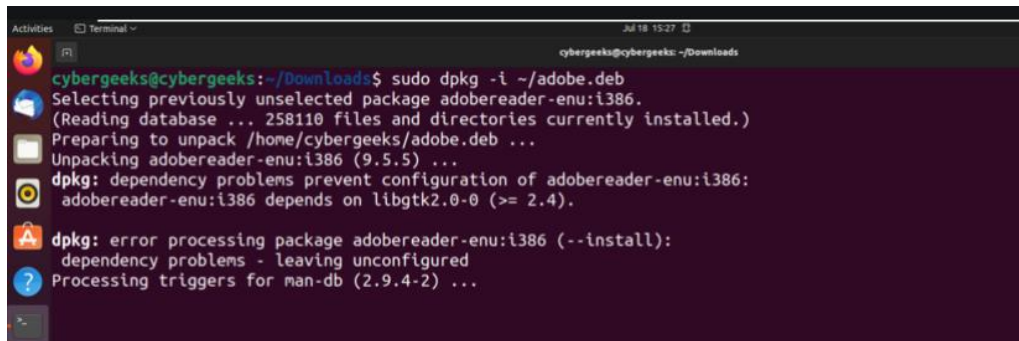


```
cybergeeks@cybergeeks: ~/Downloads
~/Downloads$ sudo apt install libxml2:i386 libcansberra-gtk-module:i386 gtk2-engines-murrine:i386 libatk-adaptor:i386 libgdk-pixbuf-xlib-2.0-0:i386
... Done
tree... Done
ation... Done
es were automatically installed and are no longer required:
```

Fig.24

→ This installs the prerequisite software required for our adobe to work

- `sudo dpkg -i ~/adobe.deb`



```
cybergeeks@cybergeeks: ~/Downloads
cybergeeks@cybergeeks:~/Downloads$ sudo dpkg -i ~/adobe.deb
Selecting previously unselected package adobereader-enu:i386.
(Reading database ... 258110 files and directories currently installed.)
Preparing to unpack /home/cybergeeks/adobe.deb ...
Unpacking adobereader-enu:i386 (9.5.5) ...
dpkg: dependency problems prevent configuration of adobereader-enu:i386:
 adobereader-enu:i386 depends on libgtk2.0-0 (>= 2.4).

dpkg: error processing package adobereader-enu:i386 (--install):
 dependency problems - leaving unconfigured
Processing triggers for man-db (2.9.4-2) ...
```

Fig.25

→ This installs the deb package of adobe acrobat

2. Adobe is installed



Fig.26

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